

Title: Smart Plant Watering System



Course Title: Microprocessors and Microcontrollers Lab

Course Code: CSE 316

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Segment 1:

Project Title: Smart Plant Watering System

Project Subtitle: An Automated IoT-Inspired Plant Care Solution

Project Goals:

- Automate plant watering to save time and effort.
- Reduce water waste through smart decision-making.
- Monitor soil and environmental conditions to maintain healthy plant growth.
- Provide reminders for fertilizer use and plant care.

Project Description:

The Smart Plant Watering System is an Arduino-based project that helps take care of plants by giving them water only when they need it. A soil moisture sensor checks if the soil is dry, and the system turns on a small pump to water the plant. This prevents both overwatering and underwatering. An OLED display shows live data like soil moisture, temperature, and humidity. The system also gives fertilizer reminders using a real-time clock (RTC). It is a simple and useful solution for homes, offices, gardens, and small farms.

Segment 2: Project Overview

Project Overview:

The Smart Plant Watering System is designed to automate plant care by monitoring soil moisture and environmental conditions to ensure plants receive the right amount of water at the right time. The system is built on an **Arduino platform**, integrating sensors, a water pump, an RTC, and an OLED display to deliver real-time monitoring and automated control. By combining smart decision-making and reminders, the project reduces water waste, improves plant health, and makes plant care easier for users.

This technology can be applied in home gardening, rooftop farming, greenhouses, and small-scale agriculture, where it can save time, conserve resources, and help even busy users maintain healthy plants.

Functional Description:

- The soil moisture sensor detects dryness levels.
- The Arduino Uno processes this data along with environmental readings.
- If the soil is dry and no rain is expected, the relay activates the pump.
- The RTC maintains schedules, and the OLED shows system status.

Operational Workflow:

1. Sensors detect soil and environmental conditions.
2. Data is processed by Arduino.
3. Warning when the tank water runs out
4. Smart decision logic checks dryness + rain conditions.
5. Relay activates pump if watering is needed.
6. Data is displayed on 16x2 LCD Display screen.
7. Logs track watering history and reminders.

Key Features:

- Automated watering based on soil and weather data.
- Fertilizer reminder system.
- OLED display for easy monitoring.
- Scalable solution for multiple plants.

Segment 3: Module Description

Required Modules:

1. Main Controller: Arduino Uno central control system.
2. Sensors:
 - Soil Moisture Sensor
 - DHT11/DHT22
 - Rain Sensor (optional)
3. Actuators:
 - Mini Water Pump
 - Relay Module / MOSFET

4. Power Supply:

- 9V–12V Adapter or Battery
- Wires, switch

5. Additional Modules:

- RTC Module (DS3231/DS1307)
- OLED Display

Segment 4: Sensor Specifications

Segment 5: Power Supply Modules

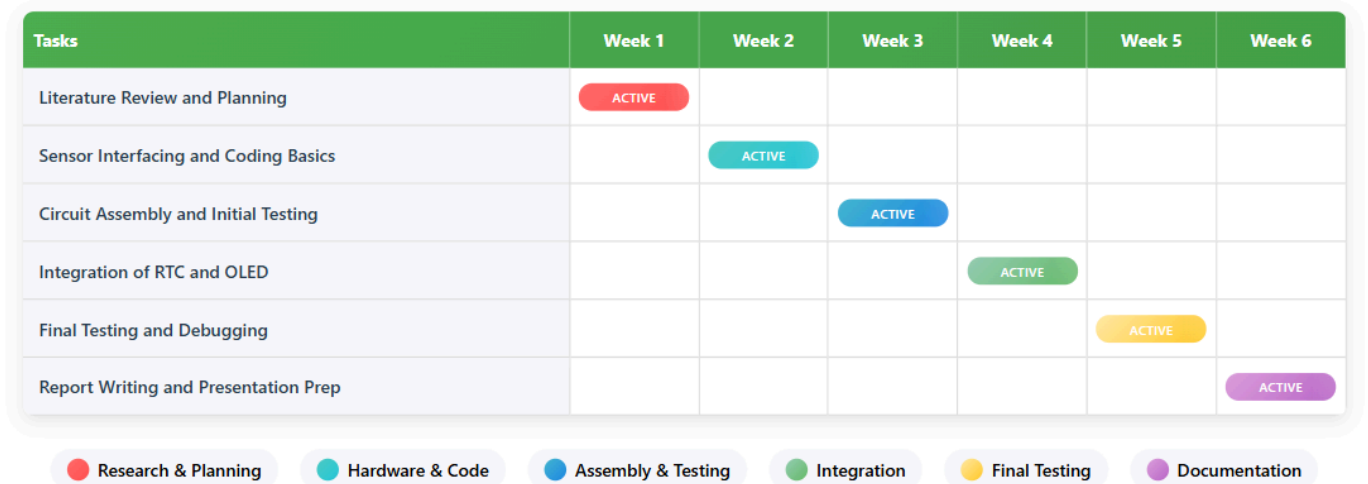
Segment 6: Circuit Diagram & Design Tools

Segment 7: Project Development Methodology

Segment 8: Project Timeline (Gantt Chart)


Smart Plant Watering System
 6-Week Development Timeline | From Concept to Implementation

Project Gantt Chart



Segment 9: Project Budget

Serial	Component	Quantity	Unit Price (BDT)
1	Arduino Uno	1	1100
2	Soil Moisture Sensor	2	500
3	DHT11 Sensor	1	200
4	Relay Module	1	400
5	Mini Water Pump	1	500

6	RTC Module DS3231	1	300
7	Float Switch	1	200
8	16x2 LCD Display /	1	400
9	Jumper Wires & Connectors	1	150
10	Power Supply (12V)	1	500

Total Estimated Cost: 4250 TK

Segment 10: CEP Mapping

1. Knowledge Profile (K's) Mapping

K's	Attributes	How K's Are Addressed Through Our Project	CO	PO
K4	Specialist Knowledge	Applied knowledge of Arduino programming, sensor data acquisition (soil moisture, DHT11), real-time control of pump and relay.	CO1, CO2	PO1
K5	Engineering Design	Designed circuit layout connecting sensors, pump, RTC, and display; developed logic for automatic watering and reminders.	CO3, CO4	PO3, PO5
K6	Engineering Practice	Built prototype using Arduino Uno, sensors, and relay; followed safe wiring, calibration, and testing for reliable operation.	CO4, CO5	PO4, PO5, PO6
K7	Comprehension	Ensured user-friendly display and app data (soil %, temp, pump status); system calibrated for accuracy and easy use.	CO5	PO7

2. Complex Engineering Problems (P's) Mapping

P's	Attributes	How P's Are Addressed Through Our Project	CO	PO
P1	Depth of Knowledge Required	Required understanding of soil moisture sensing, microcontroller coding, relay/pump control, power supply, and RTC.	CO1, CO2, CO4, CO5	PO1, PO2, PO7
P3	Depth of Analysis Required	Involved analyzing moisture thresholds, pump timing, RTC scheduling, and water-saving logic with weather data.	CO1, CO2, CO3	PO1, PO2, PO6
P7	Interdependence	Integrated multiple modules (soil sensor, DHT11, RTC, OLED, pump, Wi-Fi) requiring modular design and testing.	CO1, CO3, CO5	PO1, PO3, PO7

3. Complex Engineering Activities (A's) Mapping

A's	Attributes	How A's Are Addressed Through Our Project	CO	PO
A1	Range of Resources	Used Arduino IDE, jumper wires, relay module, DS3231 RTC, OLED display, and pump assembly.	CO1, CO4	PO1, PO6, PO7
A4	Consequences for Society & Environment	Helps reduce water wastage, supports urban gardening, and promotes sustainable plant care with minimal human effort.	CO5, CO7	PO7

Segment 11: Conclusion & References

Conclusion:

The Smart Plant Watering System makes taking care of plants easy by watering them only when needed. It helps prevent overwatering or underwatering, saves water, and is low cost. The system is simple to use and is useful for homes, offices, and gardens. It is a good way to do smart and eco-friendly gardening.

References:

1. Arduino Official Website – <https://www.arduino.cc>
2. DHT11 Sensor Datasheet
3. DS3231 RTC Datasheet
4. Fritzing – <https://fritzing.org>
5. TinkerCAD Circuits – <https://www.tinkercad.com>
6. 16x2 LCD Display – SparkFun Electronics
7. Soil Moisture Sensor Datasheet